

Sikizwa Voice 360° System Report

5. System Overview

The Sikizwa Voice 360° ecosystem consists of four primary components: **Mobile Application** – Provides the primary user interface and voice capture functionality. **Backend API Server** – Manages authentication, business logic, data storage, and integrations. **AI Analysis Service** – Performs transcription, emotion detection, risk analysis, and conversational support. **Administrative Dashboard** – Provides monitoring and management tools for administrators and counsellors.

6. System Architecture

Architectural Components

User Device → Flutter Mobile Application → Node.js Backend API → MongoDB Database → AI Service (FastAPI) → OpenAI & Machine Learning Models → Admin Dashboard

Architecture Model

The platform follows a microservice architecture where AI workloads are isolated from core business operations for improved scalability and maintainability.

7. Technology Stack

Mobile Layer: Flutter, Dart

Backend Layer: Node.js, Express.js

Database Layer: MongoDB

Caching Layer: Redis

Real-Time Communication: Socket.IO

AI Layer: Python, FastAPI, OpenAI APIs, Hugging Face Transformers, Whisper Speech Recognition

Dashboard Layer: React, TypeScript, Vite

Infrastructure: Docker, Ubuntu Server, Nginx

8. Functional Requirements

User Management: Registration, Authentication, Profile Management

Voice Processing: Voice Recording, Audio Upload, Speech Transcription

AI Analysis: Emotion Detection, Risk Scoring, Distress Identification

Emergency Response: Alert Generation, Emergency Escalation, Counsellor Notification

Reporting: Analytics Dashboard, Incident Reports, Performance Metrics

9. Non-Functional Requirements

Performance: Response time below 3 seconds, Real-time alert generation.

Scalability: Support for increasing users, Horizontal scaling capability.

Reliability: High system availability, Fault tolerance mechanisms.

Security: Secure authentication, Data encryption, Access control.

10. System Components

Mobile Application: Voice capture, User interaction, Notification handling, Device pairing.

Backend Server: API management, Authentication, Data persistence, Service orchestration.

AI Service: Speech-to-text conversion, Emotion classification, Risk assessment, Conversational AI.

Dashboard: Monitoring, User management, Report generation, Emergency tracking.

11. AI Processing Framework

Speech Transcription: Audio input is processed using Whisper-based speech recognition models.

Emotion Analysis: Happiness, Sadness, Fear, Anger, Anxiety.

Risk Assessment: Emotional indicators, Language patterns, Distress keywords.

Conversational AI: Provides empathetic responses and support recommendations.

12. Database Design

Users, Counsellors, Reports, Distress Signals, Emergencies, Notifications.

13. Security Architecture

Authentication: JWT Authentication, Secure Session Management.

Protection Mechanisms: CSRF Protection, Rate Limiting, Input Validation, XSS Prevention.

Data Protection: Encrypted Communications, Secure Storage, Access Controls.

14. Deployment Architecture

Internet → Nginx Reverse Proxy → Backend API → MongoDB + Redis → AI Service → Admin Dashboard

15. System Workflow

1. User records voice input.
2. Audio is uploaded to the backend.
3. AI service transcribes speech.
4. Emotion analysis is performed.
5. Risk level is calculated.
6. Results are stored.
7. Dashboard updates are generated.
8. Emergency procedures are triggered if required.